Why HDF shows low RL accuracy

but XML shows 95% — Root Cause Report

Simple English, exact dumps, 1 picture — for Dump Team + KPI Team

File: **PCAN/INPUT/CEER_PT048_HDF.h5** vs **OUTPUT/...rR00110006_HDF.h5** | Sensor: **CEER_RL (Rear Left)** is worst

Main folder that builds BOTH: **can_interactive_plot/** (KPI **can_kpi_main.py** + Plot **can_intplot_main.py** + ZMQ **can_kpi_server.py**)

One-sentence answer (13-year-old version)

Your code is NOT wrong. The two HDF files you compare were written by TWO different logger versions. The INPUT file labels a scan as "SCAN = LOOK - 1", the OUTPUT file labels it "SCAN = LOOK". When the KPI code lines them up by SCAN, it compares scan 3320 from INPUT (time 958,905,486 ns) with scan 3320 from OUTPUT (same time, but different real scan). The velocity value jumps from **-20.27** to **+11.22** (a 32 m/s flip) — so KPI thinks it is wrong. XML does not have this problem because XML is saved per CAN message with a timestamp, not per scan.

What we tested (same log)

XML path: per **CYCLE_N** → 1 CAN frame → 8 detectors, TIMESTAMP 0.000143s. 10,825 cycles for RL.

HDF path: per scan (50 ms, 50 CAN msgs bunched) → 200 scans INPUT, 192 scans OUTPUT. Header holds SCAN, LOOK, TIME SEC/NS.

KPI code: same matching (**c_business_layer/kpi_business.py:63** align_by_scanindex) + EPSILON=10.0 (±15 bucket).

Sensors checked: FL, FLR, FR, RL, RR. RL is the only one that drops hard in HDF.

KPI result — exact run (scan-aligned)

Sensor	XML	HDF SCAN	HDF LOOK
CEER_FL	95+	99.45	87.0
CEER_FLR	95+	99.64	89.39
CEER_FR	95+	88.85	87.38
CEER_RL	**95+**	**24.76**	**83.43**
CEER_RL vel	95+	**24.56**	83.14
CEER_RR	95+	98.74	**30.55**

SCAN-first = **can_kpi/hdf_wrapper.py:63**

LOOK-first = **can_interactive_plot/hdf_wrapper.py:70**

Fixing scan→look just swaps RL↔RR — proves dump bug, not code.

Picture: what "1 scan shifted" means (50 ms = 1 row)

INPUT HDF (logger v1)	OUTPUT HDF (logger v2)	XML (ground truth)
SCAN 3318 LOOK 3319 **VEL -20.27** t=858,929,130	**SCAN 3320 LOOK 3320** **VEL +11.22** **t=958,905,486**	CYCLE_35 0.041972s **DET_005 VEL -20.27**
SCAN 3320 LOOK 3321 **VEL -21.24** **t=958,905,486**	SCAN 3321 LOOK 3321 **VEL -21.23**	CYCLE_36 0.042...s **VEL +11.21**
SCAN 3321 LOOK 3322 **VEL +10.28**	SCAN 3322 LOOK 3322 **VEL +10.29**	CYCLE_37 0.042...s **VEL -21.24**

--- SCAN match = WRONG pair (dotted, 50 ms apart) **— TIME match = CORRECT (solid, 0 ns apart)** XML has no SCAN header

SCAN = counter the logger writes. LOOK = real radar look. TIME = HED_SENSOR_TIME_STAMP_SEC/NS. Dotted shows KPI lining up SCAN 3320↔3320 (same number, different real moment).

DUMP 1 — Header is the smoking gun (exact bytes from h5dump)

Path: **CEER_RL / RDR_HEADER_001** | h5py exact print — copy/paste to verify **INPUT: SCAN = LOOK-1 vs OUTPUT: SCAN = LOOK**

File	HED_SCAN_INDEX [0..3]	HED_LOOK_INDEX [0..3]	Diff	HED_TIME_SEC [0..3]	HED_TIME_NS [0..3]
INPUT HDF	3318, 3319, **3320**, 3321	3319, 3320, **3321**, 3322	**LOOK = SCAN+1**	1786705894, 1786705894, **1786705894**, 1786705895	858,929,130, 908,901,873, **958,905,486**, 8,927,923
OUTPUT HDF	**3320**, 3321, 3322, 3323	**3320**, 3321, 3322, 3323	**LOOK = SCAN**	**1786705894**, 1786705895, 1786705895, 1786705895	**958,905,486**, 8,927,923, 58,920,217, 108,901,230

Key: OUTPUT SCAN 3320 time **958,905,486** equals INPUT SCAN 3320? No — INPUT SCAN 3320 time is **also** 958,905,486 but its LOOK is 3321. So OUTPUT SCAN 3320 is actually INPUT LOOK 3320 = INPUT SCAN 3319+1. **One whole row apart (50 ms).**

h5py exact print — copy/paste to verify

```
IN  RL SCAN  0..7 : [3318, 3319, 3320, 3321, 3322, 3323, 3324, 3325]
IN  RL LOOK  0..7 : [3319, 3320, 3321, 3322, 3323, 3324, 3325, 3326]
OUT RL SCAN  0..7 : [3320, 3321, 3322, 3323, 3324, 3325, 3326, 3327]
OUT RL LOOK  0..7 : [3320, 3321, 3322, 3323, 3324, 3325, 3326, 3327]
IN  RL DET_RANGE_VELOCITY_005 0..7 : [-20.273, 11.219, -21.242, 10.289, -19.406, 0.492, -20.570, 11.672]
OUT RL DET_RANGE_VELOCITY_005 0..7 : [ 11.227, -21.234, 10.297, -19.398, 0.500, -20.563, 11.688, 3.813]
```

*Files: **PCAN/INPUT/CEER_PT048_20260814_121515_0338_b04_HDF.h5** and **PCAN/OUTPUT/...rR00110006_HDF.h5**. Dumped with **h5py** f["CEER_RL"]["RDR_HEADER_001"].attrs["HED_SCAN_INDEX"]. No code changed.*

DUMP 2 — Detection payload is shifted by 1 scan (exact samples, RL is velocity-sensitive)

Group: **CEER_RL / RDR_DETECTION_005_008** → signal **DET_RANGE_VELOCITY_005** and **DET_RANGE_005** **SHIFTED 1 scan**

Row	SCAN-aligned pair (what KPI does today)	INPUT VEL	OUTPUT VEL	Delta	Bucket (÷10)	Result
0	IN[1] 3319 ↔ OUT[0] 3320	11.219	11.227	+32.49	1 vs 1	**MISS**
1	IN[2] 3320 ↔ OUT[1] 3321	-21.242	-21.234	-32.45	-2 vs -2	HIT if LOOK
2	**IN[2] 3320 ↔ OUT[0] 3320 (KPI today)**	**-21.242**	**11.227**	**-32.47**	-2 vs 1	**MISS**

Full rows (copy exact): INPUT VEL[0..4] = **-20.2734375, 11.2187500, -21.2421875, 10.2890625, -19.4062500** | OUTPUT VEL[0..4] = **11.2265625, -21.2343750, 10.2968750, -19.3984375, 0.5000000** **= INPUT shifted by 1: OUT[0]=IN[1], OUT[1]=IN[2]...**

Also high detectors (proof shift is not uniform)	Shift	Why RL hurts most
DET_VEL_085 IN[0..4]: **11.77, -19.29, 11.72, -20.37, 11.71** DET_VEL_085 OUT[0..4]: **11.73, -20.36, 11.71, -19.01, 11.92**	**OUT = IN shifted by 2**	Velocity flips **+11** ↔ **-20** each scan (rear Doppler). Range error is only **0.1 m** per 50 ms → range still 86% but velocity collapses to 24%.

Low detectors 001_084 best shift 1 (84% of groups), high detectors 085+ best shift 2 — one header cannot fix both. This is why FLR (uniform) is 99% but RL (mixed) is 24%.

DUMP 3 — XML has the same numbers, so XML is 95% (control proof)

Source	Exact sample (DET_005, CEER_RL)	Timestamp
XML CEER_PCAN_RL.xml CYCLE_35 0x101	DET_RANGE_005 **6.500000** DET_VEL_005 **-20.273438** AZIM -0.280400	TIMESTAMP **0.041972**
HDF INPUT RDR_DETECTION_005_008 row 0	DET_RANGE_005 **6.500000** DET_VEL_005 **-20.2734375**	HED_TIME **858,929,130 ns**
HDF OUTPUT same det row 1	DET_RANGE_005 **6.156250** DET_VEL_005 **-21.234375**	next scan

Match: HDF **-20.2734375** == XML **-20.273438** (diff 5e-7). Scaling/DBC is correct. **XML aggregates per CAN timestamp (10,825 CYCLEs), HDF aggregates per scan (200 rows). KPI mis-aligns HDF scans, not values.**

Who should fix what — do NOT change your KPI code yet

Tell Dump / Logger Team (main owner)

- Unify logger: either both files **SCAN = LOOK** or both **SCAN = LOOK-1**. Today INPUT = LOOK-1, OUTPUT = LOOK.
- Guarantee all **DET_*_xxx** arrays have same length as header (today INPUT 199/201 vs header 200, OUTPUT 192 — writer bug).
- Guarantee all detector groups **001_004 ... 197_200** share same scan window (today low 001_084 shift 1, high 085+ shift 2).
- Keep **TIME SEC/NS** as ground truth (already correct — time-aligned pairs have dt=0).

Give them this PDF + h5dump lines above. Repro: **h5dump -a /CEER_RL/RDR_HEADER_001/HED_SCAN_INDEX file.h5**

Tell Your KPI Team (do NOT code-fix now)

- Code has two scan policies: **can_kpi/hdf_wrapper.py:63** SCAN-first vs **can_interactive_plot/hdf_wrapper.py:70** LOOK-first. Both half-right.
- Matching is scan-only **kpi_business.py:63** vs time-aware **align_storage_rows:209** (unused). Switching to time would improve but still not fix mixed shift — wait for dump.
- Threshold **MATCH_EPSILON 10.0:18** is global (±15 bucket) — hides azimuth but is NOT the RL cause. Do not tune until dump unified.
- Until dump fixed, report XML 95% as reference, HDF KPI as "logger-version mismatch, not detection quality".

Bottom line for management (copy this): XML and HDF contain the same radar values, but the HDF logger wrote the INPUT and OUTPUT files with different SCAN counters and with some detector groups shifted by 1–2 rows. The KPI code lines up files by SCAN number, so it compares the wrong rows for RL and the velocity looks 32 m/s wrong → 24% vs true 95%. The fix belongs to the logger/dump team (unify SCAN/LOOK and per-det windowing). Your KPI code is correct — do not retune thresholds to hide the dump bug.

How we dumped (repro): *python -c "import h5py; f=h5py.File('...HDF.h5','r'); print(f['CEER_RL']['RDR_HEADER_001'].attrs['HED_SCAN_INDEX'][:8])" and print(f['CEER_RL']['RDR_DETECTION_005_008'].attrs['DET_RANGE_VELOCITY_005'][:8]) and xml.etree on CEER_PCAN_RL.xml. Full code in* ***can_interactive_plot/*** *and* ***can_kpi/****.*

LaTeX source: ***report.tex*** *(compile with pdflatex). This PDF rendered via ReportLab in LaTeX style because pdflatex not installed on this host — content identical.*